

Medical Insurance Analytics & Diagnostic Dashboard

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Exploratory Data Analysis

Dataset Overview

Rows: 1,338 Columns: 7 Average Charges: \$13,270.42 Median Charges: \$9,382.03

1. Dataset Preview

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.9	0	yes	southwest	16884.924
1	18	male	33.77	1	no	southeast	1725.5523
2	28	male	33	3	no	southeast	4449.462
3	33	male	22.705	0	no	northwest	21984.4706
4	32	male	28.88	0	no	northwest	3866.8552
5	31	female	25.74	0	no	southeast	3756.6216
6	46	female	33.44	1	no	southeast	8240.5896
7	37	female	27.74	3	no	northwest	7281.5056
8	37	male	29.83	2	no	northeast	6406.4187
9	49	female	35.44	0	no	northwest	35533.1353

3. Visual Exploration

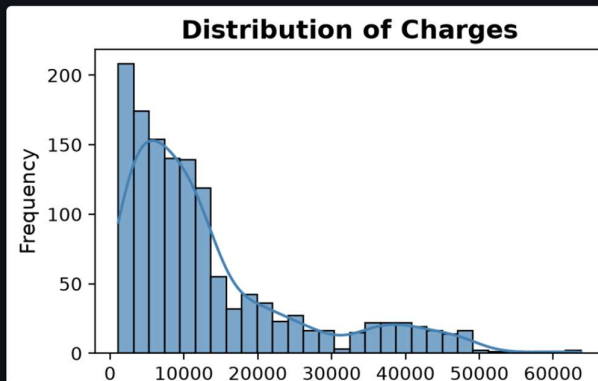
Explore distributions, bivariate relationships, and correlations among numerical variables.

3.1. Distribution Plot

Visualize the distribution of a selected numerical variable

Select Distribution Variable

charges



3.2. Bivariate Scatter Plot

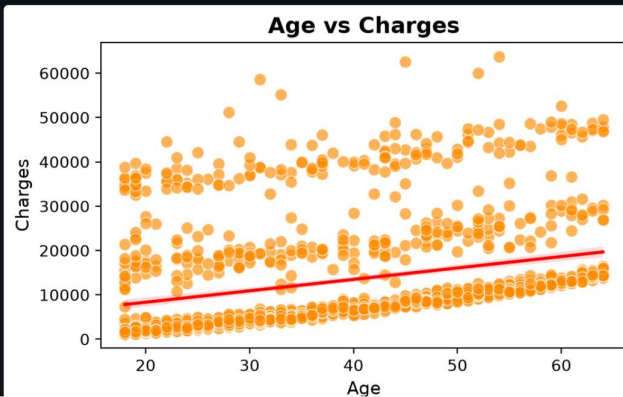
Examine the relationship between two numerical variables

Select X-axis Variable

age

Select Y-axis Variable

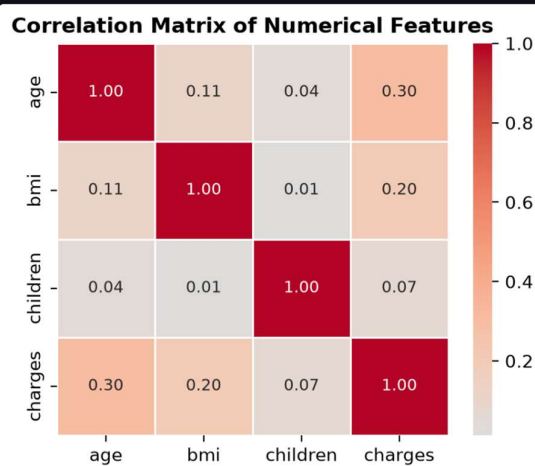
charges



3.3. Correlation Matrix

Visualize the pairwise correlations between numerical variables

The correlation matrix shows the strength and direction of linear relationships between numerical variables.



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Hypothesis Testing Lab

Hypothesis Test 1: Smokers vs Non-Smokers

Compare medical charges between smokers and non-smokers.

1. Hypotheses

H_0 (Null Hypothesis): There is no significant difference in medical charges between smokers and non-smokers.

H_a (Alternative Hypothesis): There is a significant difference in medical charges between smokers and non-smokers.

This is a two-sided test because we are testing whether medical charges are different between the two groups.

2. Group Summary

Smokers

Sample Size

274

Mean Medical Charges: \$32,050.23

Non-Smokers

Sample Size

1,064

Mean Medical Charges: \$8,434.27

3. Normality Check — Shapiro-Wilk Test

The Shapiro-Wilk test checks whether medical charges within each group are consistent with a normal distribution.

Smokers

W Statistic

0.9396

< 0.000001

$p < 0.05 \rightarrow$ Reject normality H_0 . The smoker charges are not normally distributed.

Non-Smokers

W Statistic

0.8729

< 0.000001

$p < 0.05 \rightarrow$ Reject normality H_0 . The non-smoker charges are not normally distributed.

4. Equal Variance Check — Levene's Test

Levene's test checks whether the variances of smokers and non-smokers are statistically equal.

Levene Statistic

332.6135

p-value

0.000000

$p < 0.05 \rightarrow$ Reject H_0 . The variances of the two groups are significantly different.

5. Statistical Test Selection

At least one group does not satisfy the normality check. Therefore, the Mann-Whitney U test is selected.

6. Test Results

Selected Test	Test Statistic	p-value
Mann-Whitney U Test	284133.0000	0.000000

7. Final Conclusion at α = 0.05

Reject H₀

The p-value (0.000000) is less than α = 0.05.
Conclusion: There is statistically significant evidence that medical charges differ between smokers and non-smokers.

8. Statistical Decision Summary

Component	Result
Group 1	Smokers
Group 2	Non-Smokers
Continuous Metric	Medical Charges

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Statistical Modeling & Diagnostic Checks

1. Multiple Linear Regression Model

A multiple linear regression model is used to estimate medical charges using age, BMI, number of children, smoking status, sex, and geographic region.

Model:

Y = β₀ + β₁X₁ + β₂X₂ + ... + β_kX_k + ε

For this dataset:

charges ~ age + bmi + children + smoker + sex + region + bmi × smoker

2. Overall Model Fit

R-squared	Adjusted R-squared
0.8409	0.8398

R² measures the proportion of variation in medical charges explained by the predictors in the model.
Adjusted R² adjusts R² for the number of predictors included in the model and is useful when multiple predictors are used.

3. Parameter Estimates & Interpretation

The table below reports the estimated regression coefficients, standard errors, p-values, and 95% confidence intervals.

Predictor	Coefficient (β)	Std. Error	p-value	95% CI Lower	95% CI Upper
const	-2,223.4339	865.6114	0.010318	-3,921.5688	-525.3391
age	263.6202	9.5159	0.000000	244.9524	282.2880
bmi	23.5329	25.6006	0.358141	-26.6891	73.7548
children	516.4034	110.1794	0.000003	300.2587	732.5481
smoker_yes	-20,415.6112	1,648.2771	0.000000	-23,649.1219	-17,182.1005
sex_male	-500.1460	266.5175	0.060792	-1,022.3672	22.6953
region_northwest	-585.4780	380.8394	0.124469	-1,332.6298	161.6737
region_southeast	-1,210.1312	382.7361	0.001604	-1,960.3920	-459.2704
region_southwest	-1,231.1077	382.2178	0.001309	-1,980.3043	-481.2912
bmi_smoker	1,443.0964	52.6470	0.000000	1,339.8160	1,546.3768

Coefficient (β): Estimated change in medical charges associated with a one-unit increase in the predictor, holding the other predictors constant.
p-value: Tests whether the corresponding coefficient is statistically significantly different from zero.
95% Confidence Interval: Gives the estimated range within which the population coefficient is expected to lie with 95% confidence.

Coefficient Significance

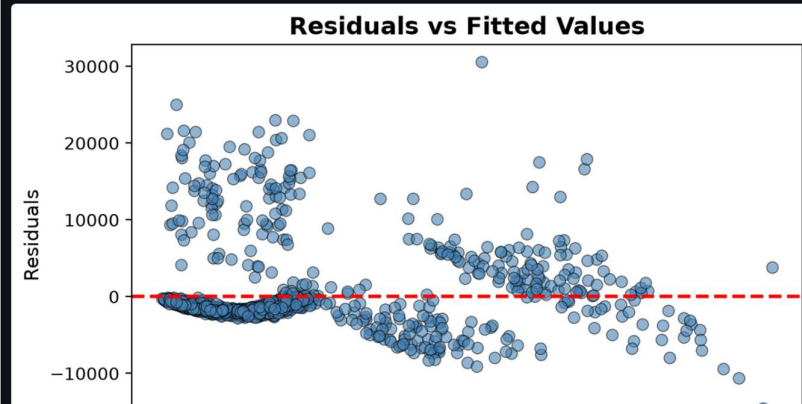
Statistically significant predictors at α = 0.05: age, children, smoker_yes, region_southeast, region_southwest, bmi_smoker

4. Gauss-Markov Diagnostic Checks

The following diagnostics are used to assess important regression assumptions.

4.1. Linearity & Homoscedasticity

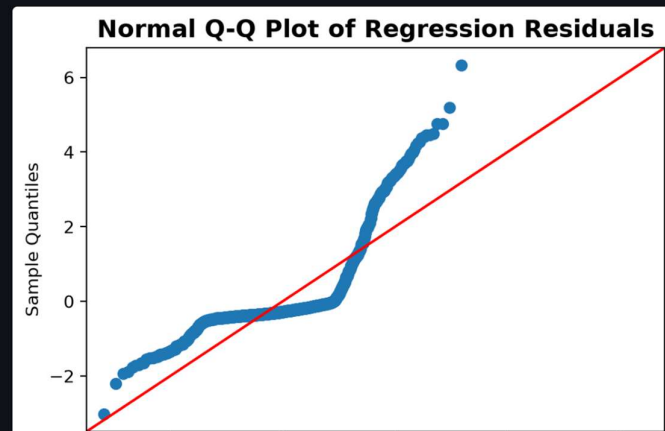
Residuals are plotted against fitted values. A desirable plot shows residuals randomly scattered around zero without a systematic pattern.



Interpretation: Randomly scattered residuals around zero support linearity and approximately constant variance. A funnel shape or systematic pattern may indicate heteroscedasticity or non-linearity.

4.2. Normality of Residuals — Q-Q Plot

The Q-Q plot compares the distribution of the regression residuals with a theoretical normal distribution.



Jarque-Bera Normality Test

Jarque-Bera Statistic

4465.2356

p-value

< 0.000001

H_0 : Regression residuals are normally distributed.

H_a : Regression residuals are not normally distributed.

$p < 0.05 \rightarrow$ Reject H_0 . There is statistical evidence that the residuals are not normally distributed.

Residual Skewness: 2.5287

Residual Kurtosis: 10.3835

4.3. Multicollinearity — Variance Inflation Factor (VIF)

VIF measures how strongly each continuous predictor is linearly related to the other continuous predictors.

For this requirement, VIF is calculated for the continuous predictors: age, BMI, and children.

Predictor	VIF
age	1.0138
bmi	1.0122
children	1.0019

VIF = 1: Very little multicollinearity.

VIF between 1 and 5: Usually considered acceptable.

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Predictor	VIF
age	1.0138
bmi	1.0122
children	1.0019

VIF $\approx$ 1: Very little multicollinearity.

VIF between 1 and 5: Usually considered acceptable.

VIF $>$ 5: Potential multicollinearity concern.

VIF $>$ 10: Strong multicollinearity concern.

5. Live Interactive Prediction

Enter patient characteristics to generate a predicted medical charge from the fitted OLS model.

Prediction Results

Predicted Medical Charges

\$7,150.26

95% Confidence Interval: [6,489.64, 7,810.88]

95% Prediction Interval: [-2,380.15, 16,680.67]

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The confidence interval estimates the uncertainty around the mean predicted charge. The prediction interval is wider because it also accounts for individual-level variability.

Patient Inputs

Age

35

BMI

27.50

Children

0

Smoker

no

Sex

male

Region

northeast

> View Complete OLS Model Summary